

Econometrics 2: Home Assignment 4

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Problem 1

1. $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

2. $f(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\left\{\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\}}$
Normalizing
of error term

3. $L(\beta) = \prod_{i=1}^n p(y_i|x) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\left\{\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\}}$

4. $\ell(\beta) = \ln(L(\beta)) = -\left\{\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\} - \ln(\sqrt{2\pi\sigma^2})$

5. $\frac{\partial \ell(\beta)}{\partial \beta_1} = \frac{-\sum (y_i - \beta_0 - \beta_1 x_i) \cdot (-x_i)}{\sigma^2}$

Maximize: $\frac{\sum x_i (y_i - \beta_0 - \beta_1 x_i)}{\sigma^2} = 0$

$\Rightarrow \frac{\sum x_i (y_i - \beta_0 - \beta_1 x_i)}{n} = 0$

This is the same as OLS / MME estimator!

As a result: $\beta_1 = \frac{\sum x_i (y_i - \bar{y})}{\sum x_i (x_i - \bar{x})}$

Problem 2

In the following task, we work with a consumer credit risk dataset containing 30,000 credit card customers. Our goal is to identify the factors associated with the binary target variable *default payment next month* and to evaluate

the predictive performance of the resulting model.

Figure 1 shows that the target variable is strongly imbalanced: 77.9% of observations are non-defaults and only 22.1% are defaults. This pattern reflects the rarity of default events, but it also implies that a model can achieve high accuracy by always predicting non-default, while at the same having low recall.

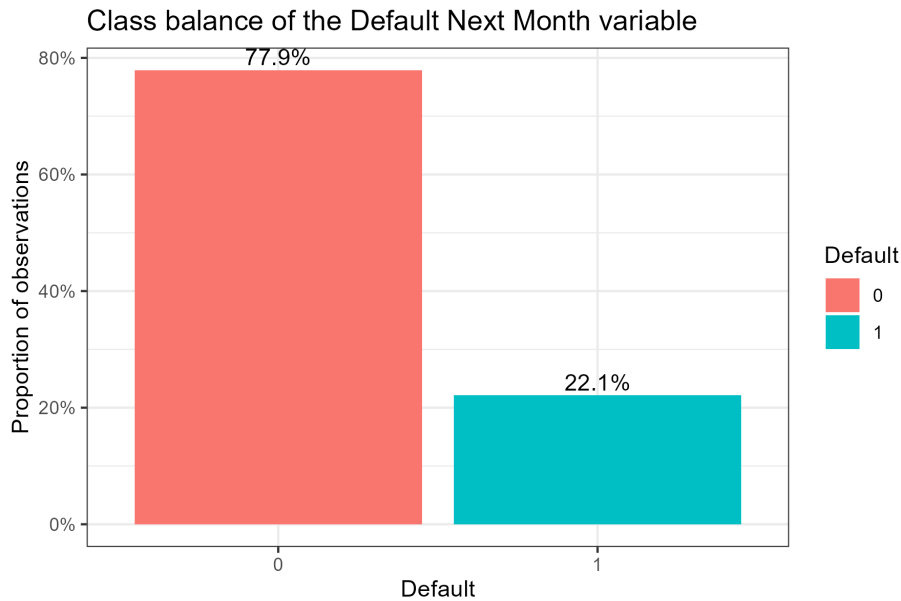


Figure 1: Class proportions of the 'Default Next Month' variable

We begin by discussing which factors may influence consumer default behaviour to justify our choice of explanatory variables on theoretical grounds.

- **Credit limit:** this variable serves as a proxy for the bank's assessment of a client's creditworthiness. Granting a high limit to risky customers would increase the share of defaults and reduce bank revenues. Therefore, a higher credit limit should, on average, be associated with customers whom the bank perceives as more reliable, and we expect it to be negatively related to default probability.
- **Sex:** this variable may capture systematic differences in behaviour between male and female clients. Some psychological evidence suggests average differences in traits such as conscientiousness¹ and risk aversion.² However, these differences may be small relative to the within-group variability. Hence, we do not expect *sex* to be among the most important predictors of default, even if it may have a statistically detectable effect.
- **Education:** this variable is likely to be informative. Individuals with higher education levels may exhibit more discipline and self-organisation (necessary to complete university studies), which can foster more regular repayment behaviour. Education is also associated with better financial literacy and critical thinking, making individuals more aware of the consequences of defaulting. Lastly, this effect generally could be due to the increase *age* variable: as people grow, they exhibit more cautiousness. We therefore might see higher education levels to be associated with a lower probability of default.
- **Marriage:** Married people should be more risk-averse as they have a family to provide for, so they should score less in openness due to their responsibilities. The effect might be ambiguous because more people in the family might lead to more expenses, but we believe people would tend to be slightly forward-looking so would avoid jeopardizing financial health of their family.
- **Age:** We expect that age can be related to financial behaviour and stability. Younger clients may have lower and more volatile income, less experience with managing credit, and therefore might face a higher probability of default. Older clients usually have more stable jobs, more savings, and experience with planning expenses, which can help them to pay in time.

¹Women are sometimes found to score slightly higher on conscientiousness, which could translate into more disciplined payment behaviour.

²Men are often assumed to be more risk-loving on average, which might be associated with a higher willingness to delay or gamble with repayments.

- **Repayment status variables:** these variables summarise how the client repaid the credit in the previous months. In the original coding, larger values correspond to longer payment delays, so higher values indicate a worse repayment history and signal that the client has already experienced difficulties paying on time. We therefore expect larger repayment status values to be associated with a higher probability of default.
- **Bill amounts:** higher bill amounts indicate that the client is using a larger share of the available credit limit. If income or repayment capacity do not increase at the same time, a high and persistent outstanding balance can raise financial pressure and make a future default more likely.
- **Repayment amounts:** these variables record how much the client actually paid in each of the same six months. Larger and more regular repayments suggest that the client is actively reducing the debt and managing the credit line well, which should be associated with a lower probability of default. In contrast, very small or zero repayments, especially when bill amounts remain high, may indicate liquidity constraints or weak repayment discipline and thus a higher default risk.

We believe the last two variable can be engineered via the difference to reflect the discrepancy between the dynamics in debt and repayment amounts. It does not matter that much if one uses more credit as long as they pay for it.

Data Structure

At this point, we should process the data and clean it to make further analysis possible. Luckily, there are no missing values, so we will not spend time deleting or imputing them. On the other hand, we have problems with inconsistent data structures in our dataset. We have identified the following issues:

- Inconsistent categories in *education* and *marriage* variables ³
- Character representation of the *bill_amt1* variable
- Duplicated values
- Outliers

Inconsistent categories issue was solved by deleting the non-existing categories⁴ and collapsing the observations that were denoted differently but had the same value to one category.⁵ The number of rows after deleting inconsistent observations decreased by 401 from 30002 to 29601.⁶ Lastly, we encoded variables as text and transformed it to factors which represent a categorical variable format in R.

Table 1: Number of unique values in each qualitative variable after cleaning.

sex	education	marriage	age	default	pay_0	pay_2	pay_3	pay_4	pay_5	pay_6
2	4	3	56	2	11	11	11	11	10	10

All bill amount variables should be numeric, so we changed the format from character to numeric. The data also contained 2 duplicated values, which we deleted. Lastly, we treated outliers by capping maximum values at the 99.8th percentile (this specific value is chosen to avoid a lot variability in regressors). We did not delete them, as there are many numeric values that are very large (see boxplots in Figure ??) and might carry important information. At the same time, some of them are apparent outliers, e.g. *pay_amt* > 100 000. We might consider applying log to those variables later.

³Repayment status variables also had a category '–2' absent in data signature. However, we will interpret it as no credit card consumption according to Jose M Sallan (2023) blogpost and the fact that there are 6450 such rows in a dataset which is roughly 21% of its original size

⁴Example: *education* - '5', '6' even though there are only 4 categories

⁵Example: *education* - 'university' as a separate category, even though university is encoded as 2

⁶See Table 1. The category counts are consistent with data signature now.

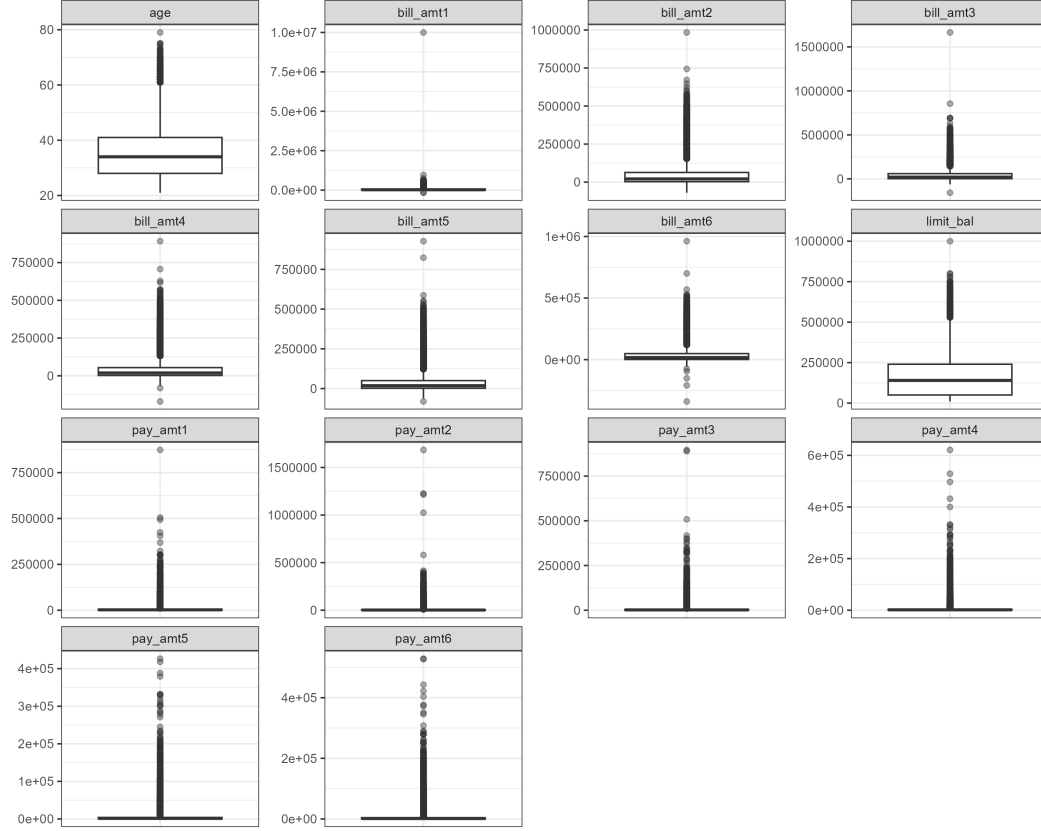


Figure 2: Boxplots of age, bill amounts and repayment amounts

Summary Statistics

After the cleaning steps, we proceed with a descriptive exploration of the continuous variables. Table 2 shows that most monetary variables (credit limit, bill amounts and repayment amounts) are strongly right-skewed: the means are noticeably higher than the medians, and the upper quartiles are far above the first quartiles. This is consistent with the idea that most clients make relatively small payments and keep modest outstanding balances, while a smaller group carries much larger debts and repayments.

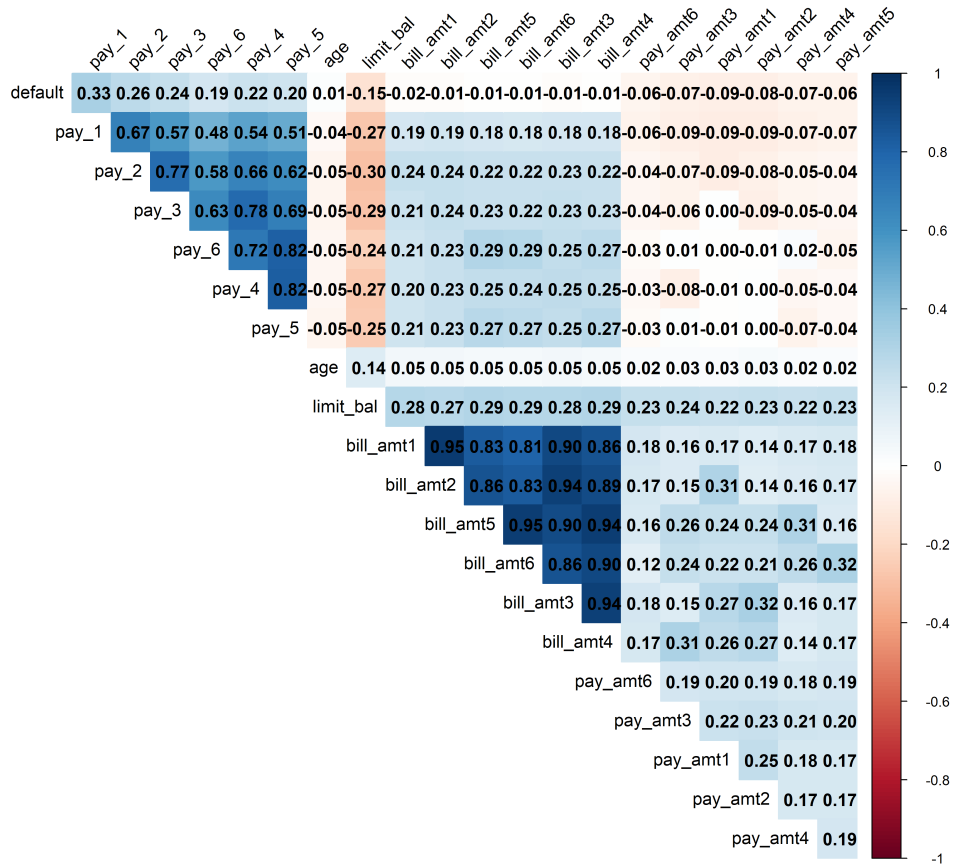
The age variable is much more concentrated: the interquartile range lies roughly between 28 and 41 years, with a minimum of 21 and a maximum of 67 (after capping). This suggests that the majority of credit card clients in our sample are in their late twenties to early forties, and that relatively few elderly customers are present in the dataset.

From the clustered correlation plot (Figure 3), we observe that variables recorded over six months (bill amounts and repayment amounts) are highly inter-correlated. This is expected, because they capture similar information about the client's credit usage and repayment behaviour within a relatively short time window. Moreover, the cluster of repayment status variables and the credit limit exhibits the highest absolute correlations with the default indicator among all numerical variables, which suggests that they may be particularly informative predictors of default.

When examining the proportion of defaults within each category in Figure 4, we see that only a small number of clients exhibit repayment statuses indicating delays longer than three months, but these categories are associated with very high default rates. In contrast, disciplined clients with statuses -2 , -1 or 0 (no use, paid in full, or paid at least the minimum amount) display substantially lower default probabilities. Among the socio-demographic variables, higher education levels are associated with lower default rates, which may partly reflect both higher financial literacy and the fact that more educated individuals tend to be older and more established in the labour market. Female customers also appear to default less frequently than male customers.

Table 2: Summary statistics of continuous variables and *age*.

Name	Min	QRT1	Median	Mean	QRT3	Max	NAs	SD
limit_bal	10000	50000.0	140000	167401.39	240000.0	620000.0	0	129336.48
age	21	28.0	34	35.46	41.0	67.0	0	9.19
bill_amt1	-165580	3528.0	22259	50858.50	66629.0	489013.2	0	72619.98
bill_amt2	-69777	2972.5	21050	48815.57	63495.5	471667.0	0	70025.56
bill_amt3	-157264	2652.0	20035	46648.28	59830.5	460205.7	0	67689.76
bill_amt4	-170000	2329.0	19006	42997.67	54273.0	434130.1	0	63264.61
bill_amt5	-81334	1780.0	18091	40100.18	50075.0	409690.8	0	59657.50
bill_amt6	-339603	1278.0	17122	38733.68	49124.5	404376.2	0	58553.64
pay_amt1	0	1000.0	2100	5468.04	5005.0	164499.9	0	13296.07
pay_amt2	0	825.5	2007	5591.46	5000.0	191667.0	0	15051.27
pay_amt3	0	390.0	1800	4983.68	4500.0	160473.5	0	13400.53
pay_amt4	0	298.0	1500	4676.96	4014.5	161031.3	0	13040.51
pay_amt5	0	259.0	1500	4658.63	4045.5	162820.1	0	13049.78
pay_amt6	0	138.0	1500	5052.06	4000.0	189482.4	0	15588.11

Figure 3: Correlation of all numeric variables and *repayment status* group.

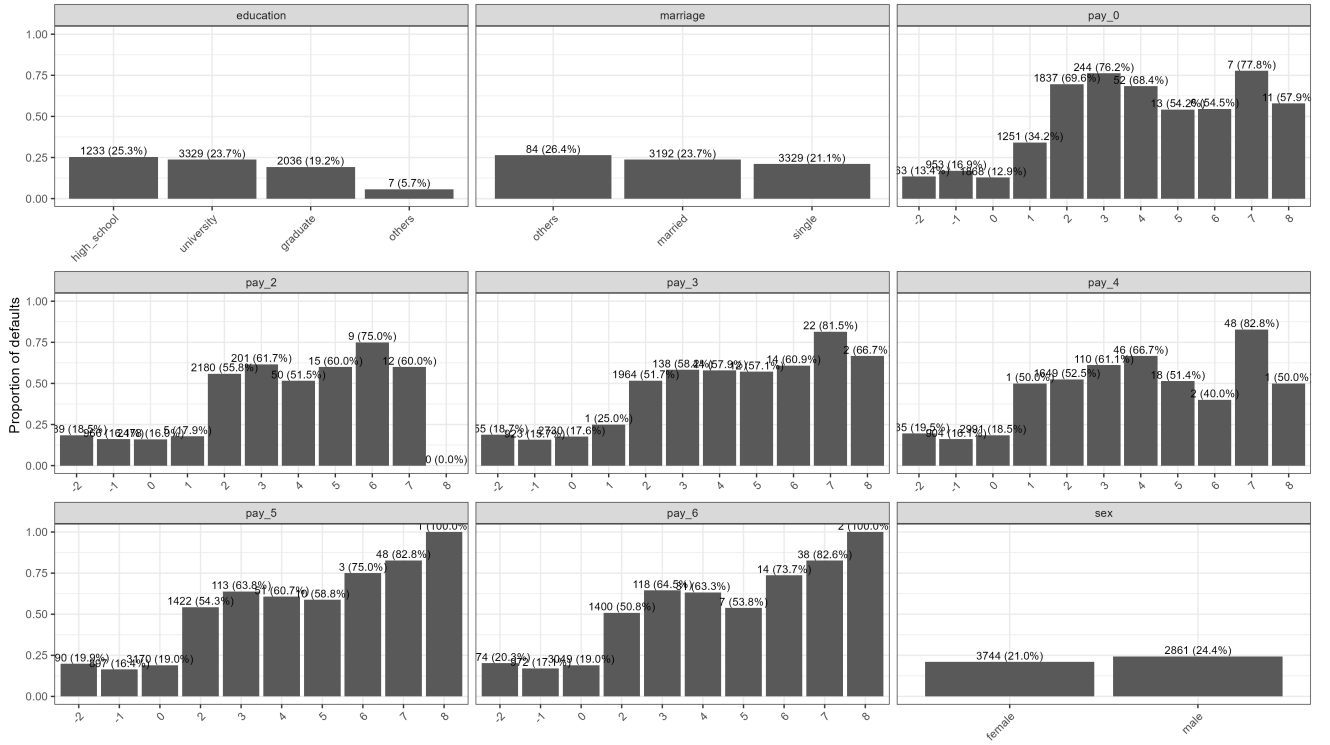


Figure 4: Share of defaults per category of qualitative variables.

These findings motivate a less detailed representation of the original variables to maximise interpretability but avoids losing explanatory power.

The following list will describe feature engineering decisions that will make our model weaker in predictive power but more interpretable

- **Repayment status variables:** They have 11 categories and 6 variables related to the performance in different month. We decide to compress 11 to 4 categories: not using credit card (-2), paid duly (-1), paid minimum (0), delay 1 or 2 months (1, 2), delay 3 and more (3+). We make this separation based on similarity of individuals within categories. First three are categories which use the credit card properly while others delay payments and split into 2 categories based on severity of delay. Afterwards, to aggregate the info between variables, we count the occurrence of each category for client and present cardinal count variables. Thus we compress 6 highly correlated variables into 4 less correlated with better interpretability.
- **Bill amounts, Repayment amounts:** those variables span 6 months so we have to compress them as well. We select summary statistics for that: median, standard deviation and maximum. Median is the most representative feature telling what bills person typically gets or how they pay for their debt. Standard deviation shows their consistency, i.e. higher spread means they do not have stability in their payments. Lastly, maximum tells to what degree they use up their credit capacity or repay it.
- **Credit utilization ratio:** this is percentage of available credit you are currently using. Combines one median bill amount with balance limit, hinting how close the person to utilizing all funds they have available.

Given that we have binary dependent variable and our goal of finding the factors it is influenced by and the magnitude of their effect, we will from select family of generalized linear models. Two models can help us: logistic regression and probit regression.

Logistic regression has a computational advantage since it is a linear regression with a sigmoid activation ($\frac{\exp(z)}{1+\exp(z)}$) function applied to it while the probit's model activation function is the normal cdf which involves an integral in computations, thus making the latter more difficult to calculate. On the other hand, probit model fits data in the economic context because of the assumption about the normality of standard errors. Both models cannot be interpreted that easily due to the inherent non-linearity, so we will deal with that later in the text.

For the main equation we will consider the specification that you can see in the Equation 1. We include *sex* dummy since there is meaningful difference between categories (there is less men but proportion of defaulted is

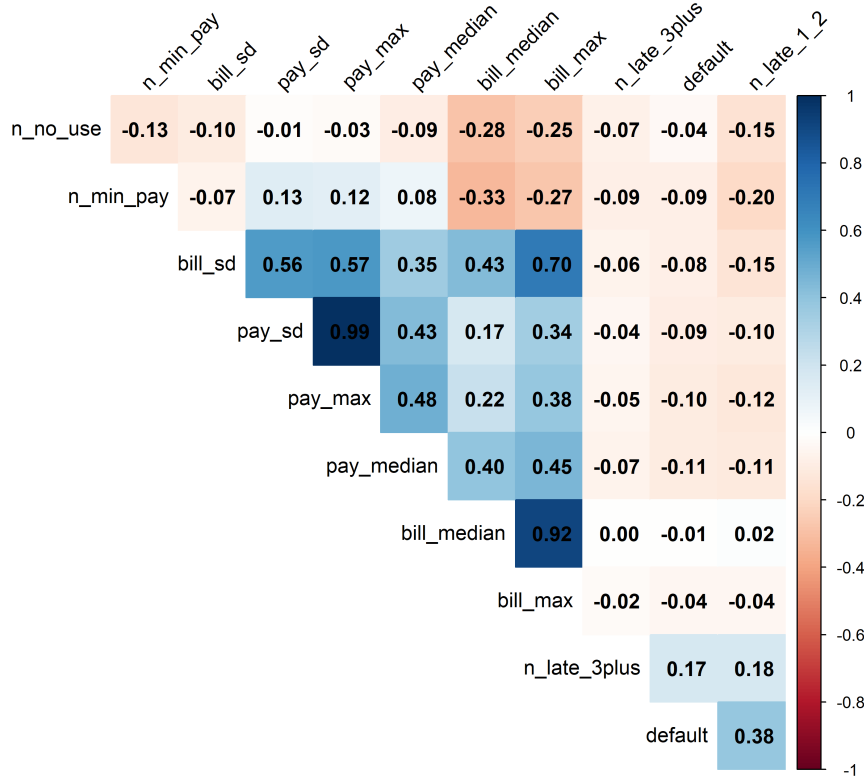


Figure 5: Correlation of engineered variables.

larger by 3.4 pp). The other variable from the main dataset will be pay_1 , i.e. payment status in April. This seems to be the most telling from all repayment status variables with correlation 0.33. Then, we added engineered credit utilization ratio, aggregated payment status variables, and repayment amount median as they had the strongest signals and lowest multicollinearity from all according to visual check of Figure 5.

From the signs of the expectation, we will discuss only **credit_util_rat_median** as we touched upon all other variables when discussing dataset. The higher credit utilization ratio might show increase in probability of default since the person approaches is credit limit and does not pay it off.

$$\begin{aligned}
 default_i = & \beta_0 + \beta_1 sex_i + \beta_2 pay_median_i + \beta_3 cred_util_rat_median_i \\
 & + \beta_4 pay_1_i + \beta_5 n_late_1_2_i + \beta_6 n_late_3plus_i + \beta_7 n_min_pay_i + u_i
 \end{aligned} \tag{1}$$

The other specification we test (Equation 2) removes engineered repayment status variables, credit utilization ratio and adds on maximum bill amount.⁷ We want to see whether feature engineering was beneficial and how important aggregated repayment status variables are.

$$\begin{aligned}
 default_i = & \beta_0 + \beta_1 limit_bal_i + \beta_2 sex_i + \beta_3 pay_median_i \\
 & + \beta_4 pay_sd_i + \beta_5 pay_1_i + \beta_6 bill_max + u_i
 \end{aligned} \tag{2}$$

⁷We leave pay_1 to have raw signal from strongest repayment status variable, but we want to see how well the aggregated variables: status and ratio assist us.

Empirical Results

Table 3: Logit and probit estimates for default

	<i>Dependent variable:</i>			
	Default (1 = default)			
	<i>logistic</i> Logit main	<i>probit</i> Probit main	<i>logistic</i> Logit 2	<i>probit</i> Probit 2
limit_bal			−0.00000 (0.00000) p = 0.000***	−0.00000 (0.00000) p = 0.000***
sexmale	0.128 (0.032) p = 0.00005***	0.075 (0.018) p = 0.00003***	0.152 (0.031) p = 0.00001***	0.088 (0.018) p = 0.00000***
pay_median	−0.0001 (0.00001) p = 0.000***	−0.00004 (0.00000) p = 0.000***	−0.0001 (0.00001) p = 0.000***	−0.00003 (0.00000) p = 0.000***
cred_util_rat_median	0.664 (0.050) p = 0.000***	0.375 (0.029) p = 0.000***		
pay_sd			−0.00001 (0.00000) p = 0.000***	−0.00001 (0.00000) p = 0.000***
pay_1delay_3plus	0.596 (0.134) p = 0.00001***	0.371 (0.081) p = 0.00001***	0.796 (0.109) p = 0.000***	0.502 (0.065) p = 0.000***
pay_1min_pay	−1.120 (0.049) p = 0.000***	−0.651 (0.029) p = 0.000***	−1.905 (0.037) p = 0.000***	−1.126 (0.021) p = 0.000***
pay_1no_use	−0.697 (0.072) p = 0.000***	−0.403 (0.041) p = 0.000***	−1.318 (0.066) p = 0.000***	−0.789 (0.037) p = 0.000***
pay_1paid_duly	−0.515 (0.055) p = 0.000***	−0.312 (0.032) p = 0.000***	−1.163 (0.047) p = 0.000***	−0.721 (0.027) p = 0.000***
n_late_1_2	0.303 (0.014) p = 0.000***	0.186 (0.008) p = 0.000***		
n_late_3plus	0.165 (0.035) p = 0.00001***	0.104 (0.021) p = 0.00001***		
bill_max			0.00000 (0.00000) p = 0.000***	0.00000 (0.00000) p = 0.000***
Constant	−1.062 (0.047) p = 0.000***	−0.654 (0.028) p = 0.000***	0.206 (0.034) p = 0.000***	0.108 (0.020) p = 0.00000***
Pseudo R^2	0.157	0.157	0.142	0.141
Accuracy	0.805	0.805	0.802	0.802
AIC	26513.3	26499.8	26982.4	27010.1
BIC	26596.3	26582.7	27065.3	27093.1
Observations	29,599	29,599	29,599	29,599
Residual Deviance (df = 29589)	26,493.350	26,479.780	26,962.380	26,990.140
Null Deviance (df = 29598)	31,426.160	31,426.160	31,426.160	31,426.160

Note:

*p<0.1; **p<0.05; ***p<0.01

Across specifications, the “main” models (Logit main / Probit main) provide a clearly better fit than the alternative specification (Logit 2 / Probit 2). The pseudo R^2 increases from about 0.141–0.142 in Model 2 to 0.157 in the main models (approximately +1.5 percentage points). This improvement is also reflected in information criteria: relative to Model 2, the main models reduce AIC by roughly 470 points and BIC by roughly 460–490 points. Since both AIC and BIC penalize additional parameters, these large declines indicate that the engineered variables in the main specification add substantial explanatory power that outweighs the cost of extra complexity. Finally, accuracy is very similar across models (about 0.80), which is not very informative under class imbalance: a model can achieve high accuracy by predicting the majority (non-default) class well while still missing many defaults. Improving the identification of defaults typically requires moving away from the default 0.5 threshold and choosing a cutoff based on the different metric than accuracy, e.g. macro-F1, a metric that penalizes missed defaults (false negatives) more than incorrect non-defaults, false positives.

Regarding coefficient direction and statistical significance, most betas are highly significant (typically $p < 0.01$), and the direction is largely consistent with economic intuition about credit risk.

Comparing the main specifications, the probit model delivers a marginally better fit than the logit model, as indicated by slightly lower information criteria (AIC and BIC). Small differences in coefficient magnitudes are expected because the two models use different activation functions: logit applies the logistic CDF, whereas probit applies the normal CDF.

We prioritize the probit specification on theoretical grounds. Under the latent-variable interpretation, probit corresponds to the case where the unobserved error term in the latent regression is normally distributed, making the normal CDF a perfect choice for mapping the latent variable into default probabilities.

Average Partial Effects

As the next step, we calculate Average Partial Effects for the main probit model to make our results interpretable:

Table 4: Average Partial Effects for continuous variables - Main probit

Variable	APE	SE	z	p	CI_low	CI_high
cred_util_rat_median	0.0932	0.0071	13.0838	0	0.0792	0.1072
n_late_1_2	0.0462	0.0020	23.2819	0	0.0423	0.0501
n_late_3plus	0.0258	0.0053	4.8640	0.000001	0.0154	0.0362
pay_median	-0.00001	0.000001	-12.9591	0	-0.00001	-0.00001

APE can be understood as the average change in default probability in percentage points *ceteris paribus*. All reported effects are statistically significant and the confidence intervals are narrow which indicates precise estimates.

In the continuous-variable Table 4, 1 pp increase in credit utilization median across 6 months raises the default probability by about 9.3 pp on average. More frequent late payments also increase default risk: each additional month in *n_late_1_2* is with about 4.6 pp, and each additional month in *n_late_3plus* with about 2.6 pp. In contrast, *pay_median* is negatively related to default, meaning that higher payments reduce default probability; however, the estimated effect is very small because the variable is measured in currency units.

In the categorical variable table 5, effects are shown as average changes between categories. For the repayment status aggregate, the baseline category is *delay_1_2*. Going to *delay_3plus* increases the default probability by about 13.3 pp while categories reflecting better repayment behavior decrease default risk. *min_pay* decreases default probability by about 18.3 pp, *no_use* by about 12.3 pp, and *paid_duly* by about 9.8 pp, all compared to *delay_1_2*. Finally, males have a slightly higher default probability on average, by about 1.9 pp.

Table 5: Average Partial Effects for discrete variables - Main probit

Var_name	Comparison	APE	SE	z	p	CI_low	CI_high
pay_1	delay_3plus - delay_1_2	0.1332	0.0300	4.4330	0.00001	0.0743	0.1921
pay_1	min_pay - delay_1_2	-0.1830	0.0091	-20.1603	0	-0.2008	-0.1652
pay_1	no_use - delay_1_2	-0.1231	0.0122	-10.1299	0	-0.1469	-0.0993
pay_1	paid_duly - delay_1_2	-0.0980	0.0103	-9.4915	0	-0.1182	-0.0778
sex	male - female	0.0187	0.0045	4.1537	0.00003	0.0099	0.0275

On the training data (Table 6), the model achieves an accuracy of 0.8050. However, this accuracy is mainly driven by good performance on the non-default class as we discussed previously. This is reflected in a high specificity (capacity of predicting class 0) of 0.9469. At the same time, the sensitivity is only 0.3108, meaning that the model correctly predicts only about 31% of actual defaults. The precision equals 0.6270, so when the model predicts default, it is correct in about 63% of cases.

The results on the test set (Table 7) are almost identical. This similarity suggests that the model does not strongly overfit, which is surprising given no regularization was applied.

Table 6: Prediction metrics for train data

Metric	Value
Accuracy	0.8050
Sensitivity (True Positive Rate)	0.3108
Specificity (True Negative Rate)	0.9469
Precision (Positive Predictive Value)	0.6270

In Table 8, we examine the confusion matrix for the test set, which confirms this pattern. Out of the actual non-default cases, the model correctly predicts 6,537 as non-default and incorrectly flags 362 as default. For the actual default cases, it correctly predicts 606 defaults but misses 1,376 defaults by classifying them as non-default.

Table 7: Prediction metrics on test set

Metric	Value
Accuracy	0.8043
Sensitivity (True Positive Rate)	0.3058
Specificity (True Negative Rate)	0.9475
Precision (Positive Predictive Value)	0.6260

Table 8: Confusion matrix for test set

	Actual 0	Actual 1
Pred 0	6,537	1,376
Pred 1	362	606

Discussion of Results and Conclusion

This study aimed at establishing key factors predicting credit default. We used logit and probit models with feature engineering to maintain decent accuracy and maximize interpretability.

The strongest predictors are repayment-status variables, especially when repayment behavior is represented by a small number of meaningful categories. This grouping is important because it preserves the key nonlinearity in credit risk while avoiding unnecessary granularity. In particular, severe delays, e.g. for of three months or more are rare, but when it occurs it is highly informative about default risk.

The second most important predictor is the credit utilization ratio measured here as the six-month median. This variable has a large APE and is economically intuitive, since it combines information about balance and limit into one indicator: how close the borrower is to exhausting available credit. From a risk-management perspective, persistently high utilization is a warning sign about default and should be monitored closely.

Differences by gender are statistically significant but relatively small in magnitude compared to repayment history and utilization. Therefore, while the sex indicator contributes to prediction, it is not a primary driver of credit risk in this dataset. Finally, the payment amount variable (*pay_median*) has an economically sensible negative association with default but is difficult to interpret in its current units; rescaling it (e.g., per thousand currency units) would make marginal effects more readable and suitable for reporting.

From the perspective of limiting expected losses, we would like to also highlight what is missing. A key limitation is the absence of direct measures of borrower capacity such as income. Income (and ideally debt-to-income measures) would allow a more structural balance limit rule, for example setting the credit limit as a multiple of income and adjusting it based on verified repayment capacity. In practice, we would treat repayment history and utilization as the main monitoring indicators after account origination. If a borrower is persistently close to the credit limit and exhibits repeated monthly delays, we would reduce exposure or tighten credit conditions. If repayment is consistently timely, risk appears low. At the same time, low utilization alone is not sufficient to classify a borrower as safe, because late-payment behavior may still signal elevated risk even when balances remain modest.

Overall, the results emphasize that repayment behavior and utilization capture the most actionable short-run signals of default, while income-based measures would likely be the most valuable addition for pre-screening.